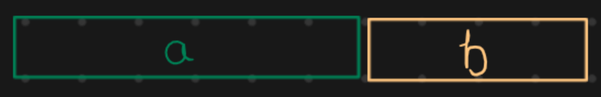
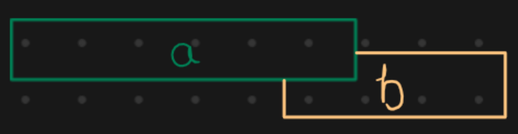
Merge Intervals

This patterns describes an efficient technique to deal with overlapping intervals In a lot of problem Involving intervals, we either need overlapping interval or merge interval if they overlap.

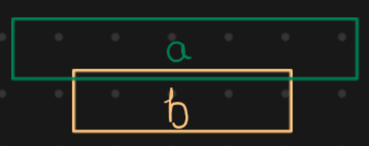
given two interval ('a' and 'b') there will be six different ways the two interval can relate to each other.

1) 

=> 'a' and 'b' do not overlap

2) 

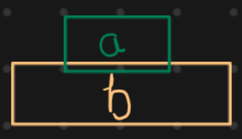
=> 'a' & 'b' overlaps, 'b' enter 'a'

3) 

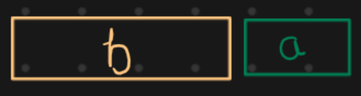
=> 'a' completely overlaps 'b'

4) 

=> 'a' and 'b' overlap and, 'a' enter 'b'

5) 

=> 'b' completely Overlaps 'A'

6) 

=> 'a' and 'b' do not Overlaps

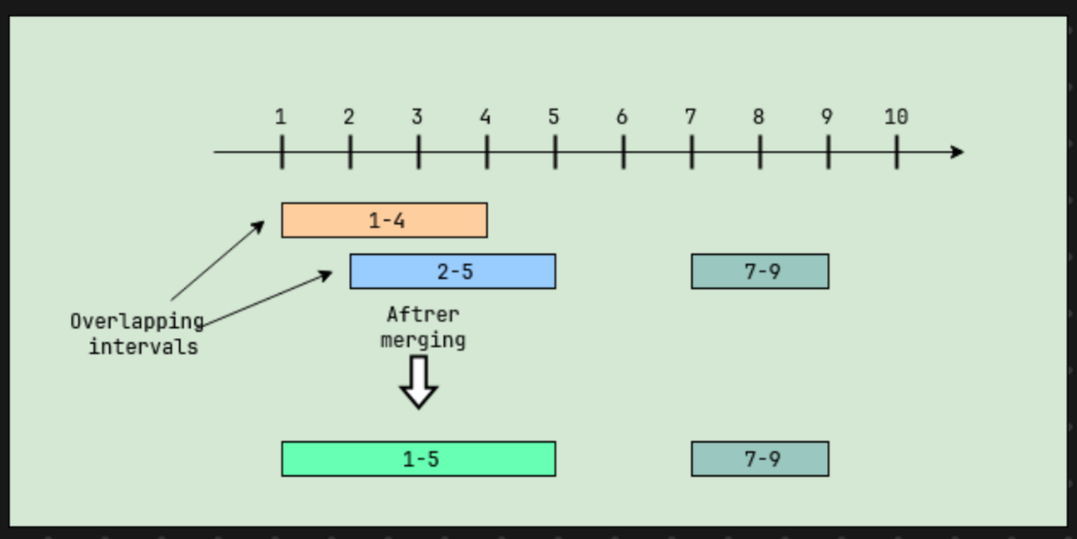
Problem - Merge Intervals

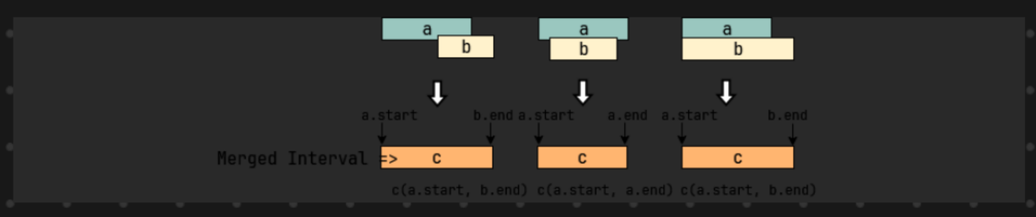
Problem Statement #

Given a list of intervals, **merge all the overlapping intervals** to produce a list that has only mutually exclusive intervals.

Example 1:

```
Intervals: [[1,4], [2,5], [7,9]]
Output: [[1,5], [7,9]]
Explanation: Since the first two intervals [1,4] and [2,5] overlap, we merged them into one [1,5].
```






```
# include <bits/stdc++.h>
using namespace std;
```

```
class Interval {
    public:
        int start = 0;
        int end = 0;
        Interval(int start, int end) {
            this -> start = start;
            this -> end = end;
        }
};

class MergeIntervals {
    public:
        static vector<Intervals> merge (vector< Interval >
            if ( interval.size() < 2) {
                return intervals;
            }
            // Sort the intervals by start time.
            sort ( intervals.begin() , intervals.end() , [ ] (const Interval &x ,
                                const Interval &y){ return x.start < y. start; });

            vector<Interval> mergedIntervals;
            vector<Interval>:: const_iterator intervalItr = interval. begin();
            Interval interval = *intervalItr++;
            int start = interval .start;
            int end = interval.end;
            while ( intervalItr != intervals.end()) {
                interval = * intervals Itr++;
                if ( interval.start <= end ) {
                    end = max ( interval.end , end);
                } else {
```

```
                mergedIntervals.push_back({start, end});
                start = interval.start;
                end = interval.end;
            }
        }
        mergedIntervals.push_back({start, end});
        return mergeIntervals;
    }
};
```

A cleaner version

```
class MergeIntervals {
public:
    static vector<Interval> merge(vector<Interval>& intervals) {
        if(intervals.size() < 2) return intervals;
        // Sort by start time
        sort(intervals.begin(), intervals.end(), [](const Interval& a, const
                                                    Interval& b){
                                                        return a.start < b.start;
                                                    });
        vector<Interval> merged;
        Interval current = intervals[0];

        for(const interval& : intervals){
            if(interval.start <= current.end){
                // Overlapping so merge
                current.end = max(current.end, interval.end);
            }else{
                // No overlap, push current and reset
                merged.push_back(current);
```

```
                current = interval;
            }
        }
        // Push the last interval
        merge.push_back(current);
        return merged;
    }
};
```

Problem - Insert Interval (medium)

Problem Statement

Given a list of non-overlapping intervals sorted by their start time, **insert a given interval at the correct position** and merge all necessary intervals to produce a list that has only mutually exclusive intervals.

Example 1:

```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,6]
Output: [[1,3], [4,7], [8,12]]
Explanation: After insertion, since [4,6] overlaps with [5,7], we merged them into on
e [4,7].
```

Example 2:

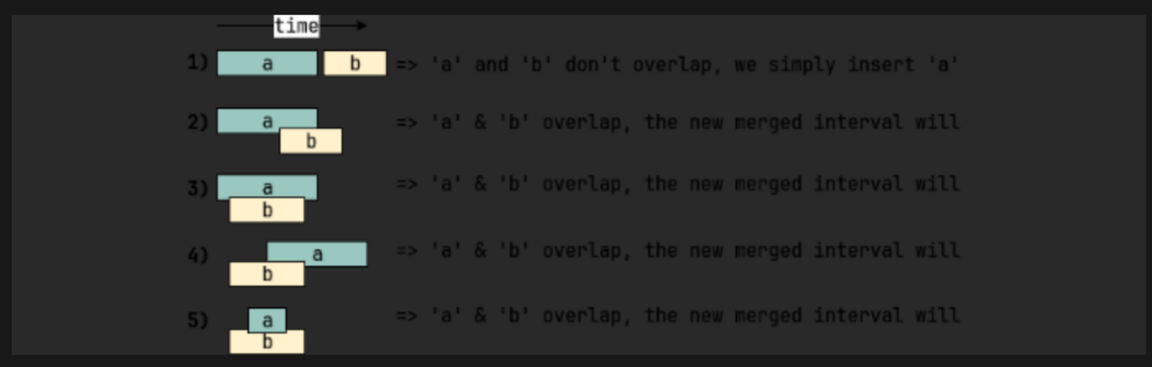
```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,10]
Output: [[1,3], [4,12]]
Explanation: After insertion, since [4,10] overlaps with [5,7] & [8,12], we merged them int
o [4,12].
```

Solution → if the given list was not sorted, we could have simply appended the new interval to it and used the merge() function from Merge Interval. Since the given list is sorted, we should try to come up with a solution better than $O(N * \log N)$.

When inserting a new interval in a sorted list, we need to first find the correct index where the new interval can be placed. In other we need to skip all the interval which end before the start of the new interval. So we can iterate through the given sorted listed of intervals and skip all the intervals with the following condition.

`intervals[i].end < newInterval.start`

Once we have found the correct place, we can follow an approch similar to Merge Interval to insert and/or merge the new interval 'a' and the first interval with the above condition 'b'. There are five possibilities:



The diagram above clearly shows the merging approach, To handle all four merging scenario

$$c.start = min(a.start, b.start)$$

$$c.end = max(a.end, b.end)$$

```
#include <bits/stdc++.h>
using namespace std;
class Interval {
 public:
    int start = 0;
    int end = 0;
    Interval(int start, int end){
        this->start = start;
        this->end = end;
    }
};

class InsertInterval {
public:
    static vector<Interval> insert(const vector<Interval>& intervals,
                                   Interval newInterval){

    if(intervals.empty()){
        return vector<Interval>{newInterval};
    }

    vector<Interval> mergedIntervals;

    int i = 0;
    // skip (and add to output) all intervals that come
before the 'newInterval'.
```

```
        while ( i < intervals . size ( ) && intervals[i].end < newInterval . start){
            mergedIntervals. push_back (intervals [i++]);
        }
        //merge all intervals that overlap with 'newInterval'
        while (i < intervals.size() && intervals[i] . start <= newInterval.end){
            newInterval . start = min( intervals[i].start , newInterval. start );
            newInterval . end = max (intervals[i] .end , newInterval . end);
            i++;
        }
        // insert the newInterval
        mergedIntervals. push_back (newInterval);
        // add all the remaining intervals to the Output
        while ( i < intervals . size ()){
            mergedIntervals . push_back ( intervals[i++]);
        }
        return mergedIntervals;
    }
};
```

Problem - Intervals Intersection

Given two lists of intervals, find the **intersection of these two lists.** Each list consists of **disjoint intervals sorted on their start time.**

Example 1:

```
Input: arr1=[[1, 3], [5, 6], [7, 9]], arr2=[[2, 3], [5, 7]]
Output: [2, 3], [5, 6], [7, 7]
Explanation: The output list contains the common intervals between the two lists.
```

Example 2:

```
Input: arr1=[[1, 3], [5, 7], [9, 12]], arr2=[[5, 10]]
Output: [5, 7], [9, 10]
Explanation: The output list contains the common intervals between the two lists.
```

Try it yourself

Try solving this question here:

```
#include <bits/stdc++.h>
using namespace std;

class Interval {
  public:
    int start = 0;
    int end = 0;
```

```
    Interval(int start, int end){
        this->start = start;
        this->end = end;
    }
};
class IntervalsIntersection {
    public:
    static vector<Interval> merge(const vector<Interval>& arr1, const vector<Interval>& arr2){
        vector<Interval> result;
        int i=0;
        int j = 0;
    while( i< arr1.size() && j < arr2.size()){
        // check if the interval arr[i] intersects with arr2[j]
        // check if one of the interval's start time lies within the other
        interval
        if((arr1[i].start >= arr2[j].start && arr1[j].start <= arr2[j].end)||
        ((arr2[j].start >= arr1[i].start && arr2[j].start<= arr1[i].end)){
            // store the intersection part
            result.push_back({max(arr1[i].start, arr2[j].start), min(arr1[i].end,
                                                    arr2[j].end)});
    }
    // move next from the interval which is finishing first
    if (arr1[i].end < arr2[j].end) {
            i++;
    } else {
            j++;
        }
    }
    return result;
    }
};
```

Problem - Conflicting Appointments

Problem Statement

Given an array of intervals representing 'N' appointments, find out if a person can **attend all the appointments**.

Example 1:

```
Appointments: [[1,4], [2,5], [7,9]]
Output: false
Explanation: Since [1,4] and [2,5] overlap, a person cannot attend both of these appointmen
ts.
```

Example 2:

```
Appointments: [[6,7], [2,4], [8,12]]
Output: true
Explanation: None of the appointments overlap, therefore a person can attend all of them.
```

Example 3:

```
Appointments: [[4,5], [2,3], [3,6]]
Output: false
Explanation: Since [4,5] and [3,6] overlap, a person cannot attend both of these appointmen
ts.
```

```cpp
#include <bits/stdc++.h>
using namespace std;
class Interval{
public:
    int start = 0;
    int end = 0;

    Interval(int start, int end){
        this->start = start;
        this->end = end;
    }
};

class ConflictingAppointments {
  public:
    static bool canAttendAllAppointments(vector<Interval>& intervals){
    // sort the intervals by start time.
    sort(intervals.begin(), intervals.end(),[](const Interval& x,
                        const Interval& y){return x.start < y.start;});

    // find any overlapping appointment
```

```
for( int i = 1; i< intervals.size(); i++) {
  if( intervals[i].start < intervals[i-1].end){
    // please note the comparison above, it is '<' & not "<="
    // while merging we needed "<=" comparison, as we will
      be merging the two intervals having condition
      intervals[i].start == intervals[i-1].end" but such
      intervals dont represent conflicting appointing as one
      starts right after the other.
    return false;
    }
  }
  return true;
 }
};
```

Problem Statement - 1

Given a list of intervals representing the start and end time of 'N' meetings, find the **minimum number of rooms** required to **hold all the meetings.**

Example 1:

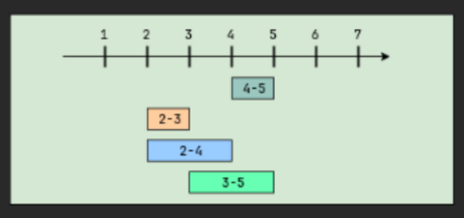
```
Meetings: [[1,4], [2,5], [7,9]]
Output: 2
Explanation: Since [1,4] and [2,5] overlap, we need two rooms to hold these two meeting
s. [7,9] can
occur in any of the two rooms later.
```

Example 2:

```
Meetings: [[6,7], [2,4], [8,12]]
Output: 1
Explanation: None of the meetings overlap, therefore we only need one room to hold all meet
ings.
```

```
Meetings: [[4,5], [2,3], [2,4], [3,5]]
Output: 2
Explanation: We will need one room for [2,3] and [3,5], and another room for [2,4] an
d [4,5].

Here is a visual representation of Example 4:
```



Solution -

Let's take the above-mentioned example

Meeting → [[4,5], [2,3], [2,4], [3,5]]

Step - 1) Sorting these meeting on thier start time will give us:
[[2,3],[2,4],[3,5],[4,5]]

Step - 2) Merging overlapping meting

- [2,3] overlap with [2,4], so after merging we'll have ⇒
  [[2,4],[3,5],[4,5]]
- [2,4] overlap with [3,5], so after merging we'll have ⇒
  [[2,5],[4,5]]
- [2,5] overlaps [4,5], so after merging we'll have ⇒ [2,5]

Since all the given meeting have merged into one big meeting ([2,5]), does this mean that they all overlapping and we need a minimum of four room to hold these meeting? you might have already guessed that the answer is No! as we can clearly see, some meeting are mutually exclusive. for example, [2,3] and [3,5] do not overlap and can happen in one room. So, to correctly solve our problem, we need to keep track of the mutual exclusiveness of the overlapping meting.

here is what our strategy will look like:

1) We will sort the meeting based on start time.
2) We will schedule the first meeting (let call it M1) in one room (lets call it r1).
3) if the next meeting m2 is not overlapping with m1, we can safely schedule it in the same room r1.
4) if the next meeting m3 is overlapping with m2 can't use r1, so we will schedule it in another room (let's call it r2).

5). Now if the next meeting m4 is overlapping with m3, we need to see if the room r1 has become free. for this, we need to keep track of the meeting happening in it. if the end time of m2 is before the start time of m4, we can use that room r1, otherwise we need to schedule m4 in another room r3.

We can conclude that we need to keep track of all the meeting currently happening so that when we try to schedule a new meeting have already ended. We need to put this information in a data structure that can easily give us the smallest ending time. A Min heap would fit our requirement best.

So our algorithm will look like this:

1. Sort all the meeting on their start time.
2. Create a min-heap to store all the active meeting, This min-heap will also be used to find the active meeting with the smallest end time.
3) Iterate through all the meeting one by one to add them in the min-heap. let say we are trying to schedule the meeting m1.
4). Since the min-heap contains all the active meeting, so before scheduling m1 we can remove all meeting from the heap that have an end time smaller than or equal to the start time of m1.
5) Now add m1 to the heap.
6) The heap will always have all the overlapping meetings, so we will need rooms for all of them. keep a counter to remember size of the heap at any time which will be the minimum number of room needed.

```
# include < bits / stdc++ .h>
using namespace std;

class Meeting {

  public:
      int start =0;
      int end = 0;
      Meeting(int start , int end){
         public:
            int start = 0;
            int end = 0;
            Meeting ( int start , int end) {
               this → start = start;
               this → end = end;
            }
};
class MinimumMeetingRooms {
   public:
      struct endCompare {
        bool operator () (const Meeting &x , const Meeting &y)
                  { return x.end > y.end; }
};

static int findMinimumMeetingRoom(vector<Meeting> & meetings){
   if( meetings. empty ()){
      return 0;
   }
   // sort the meetings by Start time
   sort ( meeting. begin ( ), meeting . end ( ),[ ] (const Meeting &x,
```

```
        const Meeting &y) {return x.start < y.start;});

    int minRooms = 0;
    priority_queue<Meeting, vector<Meeting>, endCompare> minheap;
    for(auto meeting : meetings){
        // remove all meetings that have ended.
        while(!minHeap.empty() && meeting.start >= minHeap.top().end){
            minHeap.pop();
        }
        // add the current meeting into the minHeap
        minHeap.push(meeting);
        // all active meeting are in the minHeap, so we need
        room for all of them.
        minRooms = max(minRooms,(int) minHeap.size());
    }
    return minRoom;
    }
};
```